

Lost in Instruction: Causal Sensitivity and Paraphrase Fragility in Vision-Language-Action Models

Project Defense & Presentation Slide Deck

Evaluating SmolVLA-500M, OpenVLA-7B, and OpenVLA-OFT-7.5B on LIBERO

1. Motivation: Do VLAs Truly Read Instructions?

- **Current VLA Benchmarking Deficit:** Current benchmarks test policies under static canonical prompts. High success rate does not prove language understanding.
- **Visual Shortcutting Risk:** Models can memorize visual scene layouts and execute motor skills while ignoring instruction text.
- **Core Research Goal:** Isolate the causal contribution of language by holding the physical simulator scene strictly fixed ($S_0 = \text{const}$) while systematically manipulating text prompts.

2. Benchmark & Diagnostic Probe Taxonomy

- **Scene-Fixed Causal Constraint:** $S_0(I_{\text{orig}}) \equiv S_0(I_{\text{pert}})$, matching physical state hashes across rollouts.
- **Structural & Semantic Probes:**
 - `blank` / `nonsense` $\rightarrow$ Measure baseline causal reliance (CSS metric).
 - `wrong\_object` / `wrong\_action` / `wrong\_task` $\rightarrow$ Measure Object Reliance & Verb Blindness.
- **Paraphrase Robustness (PRIDE Benchmark):**
 - `para\_object`, `para\_action`, `para\_compositional` $\rightarrow$ Measure semantic (S_K) and structural (S_T) resilience.

3. RQ1: Structural & Semantic Causal Reliance

- **Universal Object Sensitivity:** Substituting the target object noun (`wrong\_object`) drops Task Success Rate (TSR) to **7.1% across all models**.
- **Verb Ignorance Bias:** Swapping the action verb (`wrong\_action`) results in high performance retention: SmolVLA retains **57.5%**, OpenVLA retains **50.0%**, and OpenVLA-OFT retains **97.5%**.
- **Visual Shortcutting in OFT:** OpenVLA-OFT retains **12.5% success on blank instructions** due to FiLM conditioning layers.

4. RQ2: Paraphrase Fragility & PRIDE Metrics

- **Catastrophic SmolVLA Degradation:** SmolVLA-500M falls from 72.5% baseline TSR down to **4.69% overall paraphrase TSR** (PRIDE score **2.7**).
- **FiLM Conditioning Resilience:** OpenVLA-OFT-7.5B maintains **74.22% overall paraphrase TSR** (PRIDE score **65.8**).
- **Compositional Bottleneck:** Compositional paraphrases (`para_compositional`) represent the largest bottleneck across all architectures.

5. Failure Loci: Planning vs. Execution Breakdown

- **Planning Failure Dominance:** Over 56% of rollout failures under text perturbations are **Planning Failures** (gripper fails to approach/touch correct target object).
- **SmoIVLA:** 56.3% Planning Failures (897 / 1594).
- **OpenVLA:** 58.6% Planning Failures (147 / 251).
- **OpenVLA-OFT:** 58.6% Planning Failures (68 / 116).

6. RQ3: Kinematic Trajectory Divergence Profile

- **Immediate Failure Divergence:** Destructive prompts (`blank`, `nonsense`, `wrong\_object`) cause **kinematic trajectory divergence** at $t_{\text{div}} \leq 2$ steps ($e(t) > 0.05 \text{m}$).
- **Verb Perturbation Tracking:** `wrong\_action` rollouts in OpenVLA-OFT follow canonical paths for $t_{\text{div}} \approx 11.3$ steps with minimal final EEf error ($e(T) = 0.020 \text{m}$).

7. RQ4: Mechanistic Attention Allocation Ratio (AAR)

- **Early Layer Parity:** At Layer 1, Noun and Verb cross-attention are near parity ($\text{AAR} = 1.12$).
- **Exponential Verb Attention Decay:** Cross-attention to verb tokens decays rapidly across intermediate transformer layers ($l \geq 8$).
- **Extreme Late-Layer Noun Dominance:** Achieves a **12.14 times AAR** favoring object nouns in late action-generation layers, explaining verb-blindness mechanistically.

8. RQ5: Inference-Time Contrastive Guidance (ICAG)

- Training-Free Logit Modulation:

$$L_{\text{guided}}(a_t \mid O_t, I) = L(a_t \mid O_t, I) + \alpha \cdot \text{Big}(L(a_t \mid O_t, I) - L(a_t \mid O_t, I_{\text{blank}}))$$

- Performance Recovery at $\alpha = 0.5$:

- SmolVLA: Paraphrase TSR jumps from **\$4.69\%** to **28.50\%** (PRIDE score \$2.70 to 26.80\$).
- OpenVLA: Paraphrase TSR jumps from **\$44.22\%** to **68.00\%** (PRIDE score \$33.30 to 59.20\$).
- OpenVLA-OFT: Paraphrase TSR jumps from **\$74.22\%** to **88.50\%** (PRIDE score \$65.80 to 82.50\$).

9. RQ6: Long-Horizon Multi-Step Task Generalization

- **LIBERO-10 Sequential Dependency:** Multi-step tasks amplify initial language planning errors.
- **SmoIVLA:** Compositional paraphrase TSR drops from \$1.78\%\$ (Goal) to \$0.50\%\$ (LIBERO-10).
- **OpenVLA:** Compositional paraphrase TSR drops from \$24.00\%\$ (Goal) to \$10.00\%\$ (LIBERO-10).
- **OpenVLA-OFT:** Compositional paraphrase TSR drops from \$54.00\%\$ (Goal) to \$32.00\%\$ (LIBERO-10).

10. Summary of Architectural Recommendations

1. **Incorporate Asymmetric Contrastive Objectives:** Train action heads with contrastive loss against unconditioned visual features to eliminate visual shortcutting.
2. **Layer-Wise Verb Attention Boosting:** Add explicit verb-token cross-attention loss penalties to prevent verb attention decay across transformer depth.
3. **Deploy Inference-Time ICAG:** Use ICAG logit modulation ($\alpha = 0.5$) at deployment for instant paraphrase resilience without re-training.

11. Conclusion & Key Takeaways

- **Causal Isolation is Essential:** Scene-fixed causal evaluations expose visual shortcutting hidden by standard static benchmarks.
- **Mechanistic Root Cause Identified:** Verb blindness stems from late-layer cross-attention decay ($\text{AAR} = 12.14\times$).
- **Simple, Training-Free Remedy:** ICAG logit modulation effectively rescues paraphrase degradation across scales.